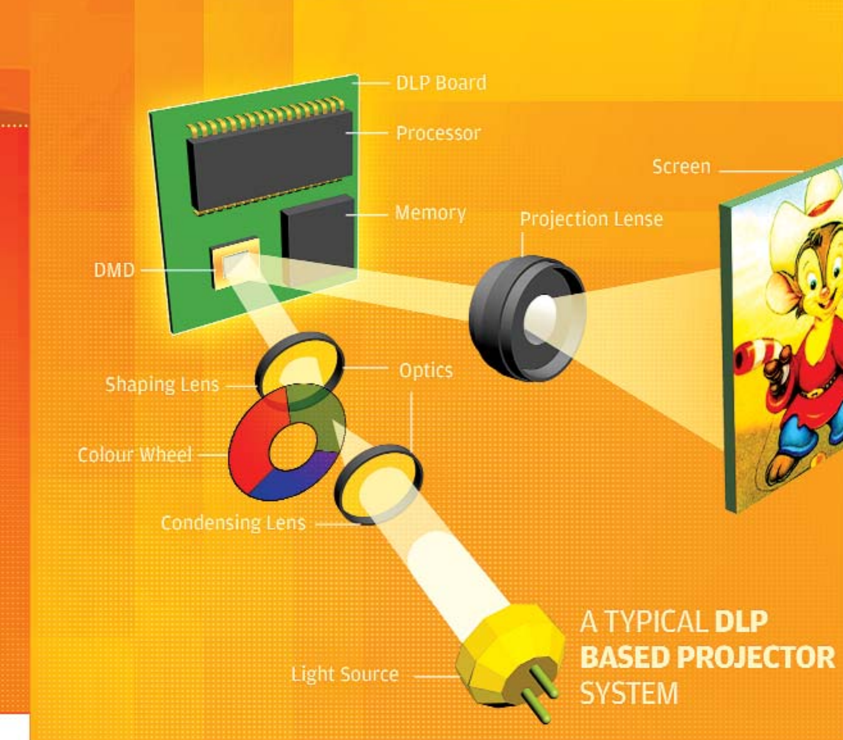


Imaging **Atul Deshmukh, Jayan Narayanan**

address pad has two pads—one feeds the address electrodes with the required signal while the other sends the 'Reset' signal to the yoke in the hinge assembly via the supporting post. The address pad acts as an interface between the mechanical mirrors and the CMOS addressing logic that forms the substrate of a DMD chip.

Since a DMD is an amalgamation of an optical device with a semiconductor controlling base, it is called an Optical Semiconductor. Such devices fall under a category known as Micro-Electronic Mechanical Systems (MEMS). A DMD is a 'MEMS' primarily because it has millions of mechanically moving micromirror's controlled via CMOS electronics.

Using a single DMD, smaller projectors can be made, which offer results on par with projectors based on slower light modulators

How Stuff Works

To modulate light in a particular direction, the mirror can be rotated to a maximum of ±12 degrees, depending on the state of the CMOS circuitry lying underneath each mirror. The CMOS applies a voltage to the address electrode of the hinge assembly to create an electrostatic attraction between the electrode and the mirror, which, in turn, results in the tilting of the mirror in the desired direction.

When the CMOS is in the 'ON' state the mirror rotates to +12 degrees, whereas when the CMOS is in 'OFF' state the mirror rotates to -12 degrees. Once the rotation completes, the mirror is electro-mechanically latched in the desired direction, and the state of the CMOS can change without affecting the position of the mirror.

Typically, for projection purposes, a metal-halide lamp is used as a light source. The white light from such a source is projected on the DMD using focusing optics.

The arrangement of the light source, with respect to the DMD chip, is such that all micro mirrors that are in the 'ON' state reflect the incident light to the pupil of the projecting lens, whereas the micromirrors in 'OFF' state reflect the incident light on to a light absorber.

The controller driving the CMOS states (ON or OFF), can change a thousand times per second. In other words, each mirror on the DMD either switches 'ON' or remains 'OFF' a thousand times per second.

When a particular micromirror remains 'ON' more frequently than 'OFF', it will project a grey pixel on the screen whereas the micromirror that remains 'OFF' than 'ON' reflects a darker shade of grey. In this way by precisely controlling the rotation of the mirrors, the DMD is able to produces a grey scale that can go up to 1024 shades of grey.

Once you have a grey scale solution in place, producing colour is just a matter of introducing a colour wheel, consisting of red, green and blue filters between the light source and the DMD.

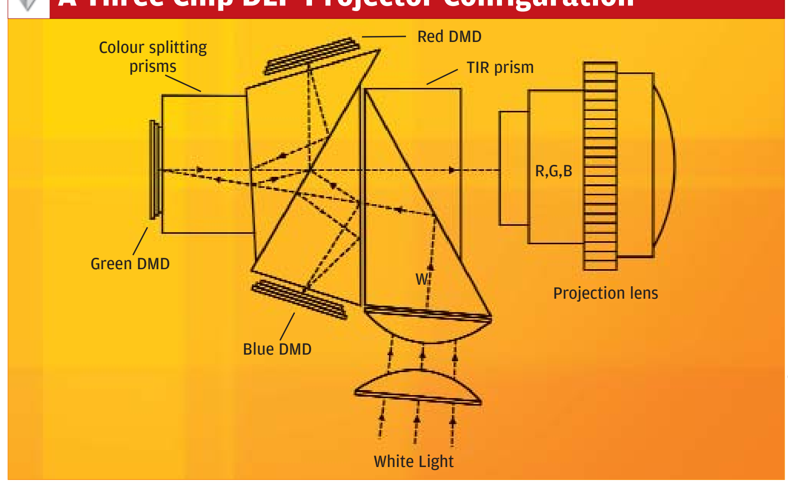
The rotation of the wheel is synchronised with the controller that drives the CMOS of each mirror. Now, for example, to project the colour purple on the screen, a pixel will only project red and blue light.

The colour green is bypassed by ensuring that the micromirror is in the 'OFF' position when the green filter is in position, and turned 'ON' when the Red and Blue filters come into position. Though this actually means that different colours are flashed on the screen a thousand times a second, the naked eye cannot keep up with this and sees a mixture of red and blue—purple.

Now, since each mirror is controlled by a digital signal—issued to the CMOS circuitry by a digital controller—the incident light is said to be digitally modulated, and hence this technique is called Digital Light Processing (DLP). It should be noted that during the whole controlling or modulating process, the incident light never undergoes an optical-to-electrical conversion, as is prevalent in digital cameras and camcorders.

The current generation of DMDs have a mechanical switching time of 15μs (microseconds) and an optical switching of 2μs. Such fast

A Three Chip DLP Projector Configuration



Infographic Sivalal S

Three DMD Chips are used in high-end DLP Projectors often used in places such as auditoriums and theatres